

Computer Science Internal Assessment

IA Project: Student Management System

Table of Contents

Criterion A - Planning	2
Defining the Problem	2
Rationale for the Proposed Solution	2
Success Criteria	3

Criterion A - Planning

Defining the Problem

Vcon International Tuitions refers to an economics tuition centre which offers tailored educational services to middle and high school students. The centre tracks grades of students, and personal data to streamline lessons accordingly. I chose this client because I am interested in education technology, and they have objective of providing quality and data-driven tutoring and making data handling process easier. The centre is presently using paper-based records and haphazardly compiled Excel sheets, which result in frequent failure like absenteeism and expired marks. Mr. Vijay Jain wastes hours in recording updates every week, and tutors have difficulties with data being organised, which slows lesson preparation and progress reports, putting their 50 students at risk of losing 10-15 percent a year.

Rationale for the Proposed Solution

In order to solve this problem I suggest creation of a desktop Student Management System (SMS) with Java as its programming language, and NetBeans as GUI and MySQL as a database storage system. This system shall enable tutors to add, store and access student records including personal details and grades using a friendly interface, whereas the owner can easily give reports. This will eliminate manual system that is error-prone with centralization database which is more accurate and efficient. Java and NetBeans were selected because of familiarity and the simplicity of creating a GUI, whereas MySQL is appropriate in organization of student data. It will allow updating records in real-time, decrease administrative load, improve planning of lessons, and increase parent satisfaction with the use of automated progress reports.

Success Criteria

1. The Student Management System will support **role-based authentication**, which will ensure **separate login credentials for Admin, Teacher and Student**.
2. **A role-based access control system** will control different menu options for the various users. Admins will get full access, while teachers and students have limited access.
3. **Unauthorised system access** will be protected through an **authentication and authorisation process**.
4. Teachers and administrators can generate student progress reports,
5. Admins can generate fee summary reports (including paid, outstanding, and overdue fees),
6. Student records are organised using **unique identifiers**,
7. The system will allow CRUD operations for each of the modules, like teachers and students.
8. The system supports **offline functionality** and is scalable to accommodate future growth in student numbers.
9. Individual teacher and student can only see their own information,
10. Valuable data (password) would be secured in the database using encryption (hash salt).
11. This system will be flexible to include the addition of existing subjects into the system to enable easy grading and mark management.
12. The system will be able to assist in printing and the generation of student grade reports with the inbuilt print facility that will have detailed data on marks.

Word count Criteria A: 246